

TUPLES & SETS

PYTHON'S SPECIAL COLLECTIONS

What are Collections in Python?

Collections are groups of items stored together in Python! Think of it like organizing your things—you can list your toys, your school books, or your friends' names all in one place. Python gives us different types of collections for different jobs. Today we'll meet two special ones: **tuples** and **sets**.

[Lists](#) → [Tuples](#) → [Sets](#)

1. What is a Tuple?

A **tuple** is a collection of items that stays in order and cannot be changed. Think of it like a train with fixed coaches—once the coaches are set, they stay in the exact same order!

Real-life Analogy: The Rajdhani Express & Aadhaar Card

The train coaches S1, S2, S3, AC1, AC2 are always in the same order. You cannot swap or rearrange them—just like a tuple! Once the train is set, the order stays the same forever.

Another example: Your **Aadhaar card** has your name, date of birth, and number that never change—just like tuple data!

Creating & Accessing Tuples

Create a tuple using standard parentheses (), with items separated by commas:

```
train_coaches = ("S1", "S2", "S3", "AC1", "AC2")
print(train_coaches)

# Use index numbers to get items - index starts at 0, not 1!
print(train_coaches[0]) # Output: S1
print(train_coaches[3]) # Output: AC1
print(len(train_coaches)) # Output: 5
```

Tuple Properties

• Ordered

• Immutable (Fixed)

• Duplicates OK

• Mixed Types

× What happens if you try to change a tuple?

```
train_coaches[0] = "B1"  
# TypeError! Python will give an error!
```

You **cannot change** a tuple once it is made. Tuples are **immutable**, which means their items stay fixed forever!

2. What is a Set?

A **set** is a collection of items that has no fixed order and absolutely no duplicates. Think of it like a **dabba (tiffin box)** with unique dals—you can have rajma, chana, and moong, but no dal appears twice!

Real-life Analogy: The Tiffin Box & School Roll Call

Imagine your school dabba has rajma, chana, and moong, and you accidentally add rajma again. In a set, the duplicate rajma disappears automatically!

Another example: During **roll call**, each student's name appears only once in the attendance register—no repeats allowed! That's exactly how sets work.

Creating Sets & Their Properties

Use curly braces { } to create a set:

```
dals = {"rajma", "chana", "moong", "rajma"}  
print(dals)  
# Output: {'rajma', 'chana', 'moong'} (Notice that the extra 'rajma' is gone!)
```

Changing Sets — Add, Remove & Check

Sets can change after creation! You can use `.add()` to put new things in and `.discard()` to take things out:

```
student_names = {"Priya", "Arjun", "Sneha"}  
student_names.add("Kiran")  
student_names.discard("Sneha")  
print(student_names) # Sneha is gone, Kiran is added!  
  
print(student_names[0])  
# TypeError! Sets are UNORDERED, so no indexing allowed!
```

Check if an item exists inside a set using the `in` keyword:

```
print("biryani" in my_foods) # Output: True
print("pizza" in my_foods)  # Output: False
```

3. Comparison: List vs Tuple vs Set

Feature	List	Tuple	Set
Symbol Used	Square Brackets []	Parentheses ()	Curly Braces { }
Ordered?	✓ Yes	✓ Yes	× No
Can Change? (Mutable)	✓ Yes	× No (Immutable)	✓ Yes
Allows Duplicates?	✓ Yes	✓ Yes	× No (Unique only)
Allows Indexing? ([0])	✓ Yes	✓ Yes	× No

4. When to Use Which?

Use a Tuple when data must stay fixed:

- Days of the week
- Your date of birth
- Fixed IPL teams for the season
- Coordinates of your school

Use a Set when you want unique items only:

- Unique states you have visited
- Languages spoken by your classmates
- Items sold in the school canteen
- Different sports played at school

5. Coding Activity — My Indian Data Box

```
# Let's try it out!
my_states = ("Maharashtra", "Kerala", "Rajasthan", "Punjab", "Goa")
print(my_states[0]) # Output: Maharashtra
# my_states[0] = "Gujarat" → This would crash because tuples are locked!

my_foods = {"idli", "dosa", "paratha", "biryani", "idli"}
my_foods.add("pani puri")
print(my_foods) # 'idli' will only appear ONCE!
```

PRACTICE WORKSHEET

TEST YOUR PYTHON POWERS! (NO ANSWERS ALLOWED)

Solve these 12 exciting questions to show how well you understand Tuples and Sets! Write your answers clearly in the space provided.

Q1. If a collection uses standard round parentheses like `my_data = ("Apple", "Mango", "Banana")`, what type of collection is it?

Q2. What special word do we use to describe a collection that can NEVER be changed once it is created? (Hint: It starts with the letter 'I').

Q3. If you create a set called `my_dabba = {"Rajma", "Chana", "Rajma", "Moong"}`, what will happen to the second "Rajma" when Python prints the set?

Q4. In Python, what is the starting index number for any ordered collection? Is it 0 or 1?

Q5. Look at this code: `toy_box = {"Car", "Doll", "Blocks"}`. If we try to run `print(toy_box[0])`, what will Python do and why?

Q6. Which Python command allows you to add a new item to an existing set? Write down the command name.

Q7. Match the bracket symbols to their correct collections:

- a) [] is for a _____
- b) () is for a _____
- c) { } is for a _____

Q8. If Rahul wants to store the names of all 7 days of the week in order so that nobody can accidentally alter or delete them, should he use a Tuple or a Set?

Q9. What keyword can you use to quickly check if the item "biryani" exists inside a set called my_foods?

Q10. True or False: A tuple can store different types of data together, like a mix of numbers and text words.

Q11. Imagine you have a list with duplicate items: [1, 2, 2, 3, 4, 4]. If you convert this list into a Set, how many total items will be left inside the set?

Q12. What specific built-in Python command or function can tell you the total number of items inside a tuple or a set? (Hint: it helps you find the length).
